

7. (a) (i) Chlorine is a non-metal found in Group 7 of the Periodic Table. When it is bubbled into a solution of potassium iodide there is a colour change from pale green to brown. Explain why this reaction occurs. [2]

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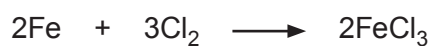
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- (ii) Write the balanced symbol equation for the reaction between chlorine and potassium iodide. [2]

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- (b) The symbol equation for the reaction between iron and chlorine is as follows.



Calculate the mass of chlorine needed to react with 1.32 g of iron. [3]

$$A_r(\text{Fe}) = 56$$

$$A_r(\text{Cl}) = 35.5$$

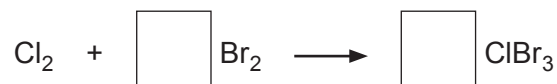
Mass of chlorine = ..... g



- (c) (i) Under certain conditions, Group 7 elements will react with each other to produce new compounds.

When chlorine is reacted with bromine, chlorine tribromide is made.

Balance the symbol equation for this reaction. [1]



- (ii) A chemist calculated that if she reacted 7.00 g of chlorine with an excess of bromine, the theoretical mass of chlorine tribromide produced is 27.55 g.

However, when she carried out the experiment using 7.00 g of chlorine the mass of chlorine tribromide obtained was 21.34 g.

Calculate the percentage yield of chlorine tribromide. [1]

Percentage yield = ..... %

